Tiago Ferrão-Lindqvist

Mechanical Engineer | Test Data Processing & Reporting

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design, testing and validation of automotive and industrial components. Develops Python tools to process test-rig data, check data against specifications and generate reports. Experienced with SQL Server queries, staging-table summaries and Power BI dashboards. Seeking to bring this practical data workflow experience into data engineering.

SKILLS

Programming and data processing: Python, pandas, NumPy, Matplotlib, npTDMS, MATLAB/Simulink
Databases and reporting: SQL, SQL Server, Power BI, Excel/VBA
Personal project tools: Flask, SQLite, Git, Docker Compose
Engineering and testing: ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, Minitab, design of experiments (DOE), GD&T, DFMEA, 8D problem solving

EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components | Setúbal | Mar 2019–Present

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare test reports for OEM customers.
- Developed Python scripts using pandas, NumPy, Matplotlib and npTDMS to process TDMS and CSV rig files (about 2 GB per week), check channels against test specifications and generate standard report figures. Reduced report preparation from about three days to half a day per test.
- Run the tool nightly on a lab PC through Windows Task Scheduler; it writes summary results to an IT-created staging table in SQL Server. Packaged as an internal tool used by four other test engineers.
- Built a Power BI dashboard over the staging table showing pass/fail trends and cycles to failure by product generation; used by about 15 validation and design engineers.
- Query the plant's SQL Server test-results database for trend analysis across three product generations.
- Correlate rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures; proposed a redesigned adjuster rail bracket that cut warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive, used to select rig load profiles that match customer duty cycles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps | Porto | Sep 2014–Feb 2019

- Designed centrifugal pump casings and impellers in SolidWorks, produced GD&T production drawings and owned DFMEA for two product lines.
- Performed structural FEA of casings in ANSYS; supported a casing-family change from grey cast iron to ductile iron that saved 12% in material cost.
- Supported conversion of the pump performance test bench to a PLC-controlled setup, changing setpoints and timers in TIA Portal under an automation engineer's supervision.

- Wrote a LabVIEW acquisition program for the performance test bench, reducing test setup time from about two hours to 30 minutes.

PROJECTS

Bike-sharing trips API | Personal learning project | 2025

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints. Runs on a personal laptop; followed a tutorial to run it with Docker Compose.

EDUCATION

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Online data engineering course | In progress
Completed 6 of 9 modules.

LANGUAGES

Portuguese (native), English (fluent), Spanish (good)